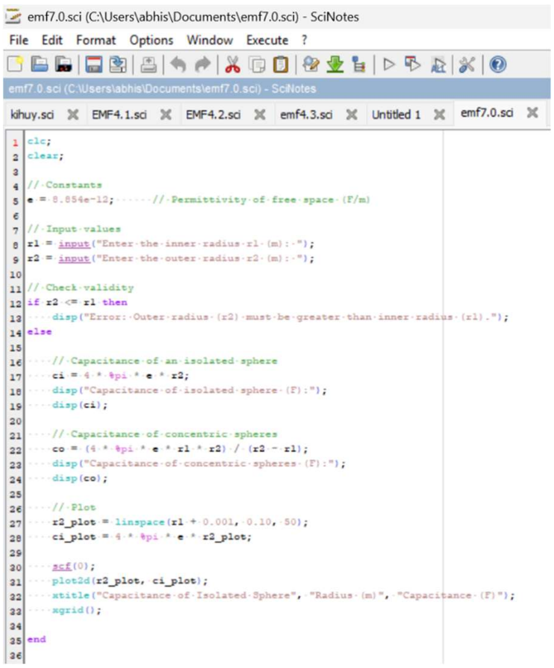
Exp:7-Determination and plotting of Capacitance of isolated and concentric sphere with different inner and outer radius and dielectric constant of the dielectric mediums.

Aim: To write program for Capacitance of an isolated sphere and two concentric sphere using SCILAB.

Software required:

•SCILAB version 6.0.1



emf7.0.sci (C:\Users\abhis\Documents\emf7.0.sci) - SciNotes

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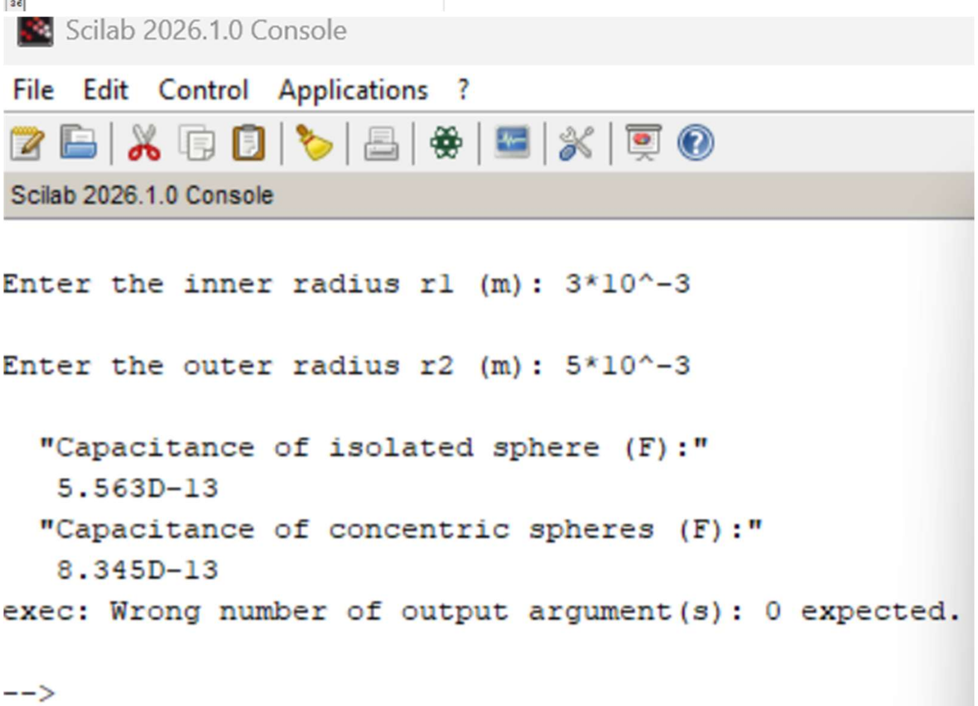
```
clc;
clear;

// Constants
e = 8.854e-12;      // Permittivity of free space (F/m)

// Input values
r1 = input("Enter the inner radius r1 (m): ");
r2 = input("Enter the outer radius r2 (m): ");

// Check validity
if r2 <= r1 then
    disp("Error: Outer radius (r2) must be greater than inner radius (r1).");
else

    // Capacitance of an isolated sphere
    ci = 4 * %pi * e * r2;
    disp("Capacitance of isolated sphere (F):");
    disp(ci);

    // Capacitance of concentric spheres
    co = (4 * %pi * e * r1 * r2) / (r2 - r1);
    disp("Capacitance of concentric spheres (F):");
    disp(co);

    // Plot
    r2_plot = linspace(r1 + 0.001, 0.10, 50);
    ci_plot = 4 * %pi * e * r2_plot;

    scf(0);
    plot2d(r2_plot, ci_plot);
    xtitle("Capacitance of Isolated Sphere", "Radius (m)", "Capacitance (F)");
    xgrid();

end
```



Scilab 2026.1.0 Console

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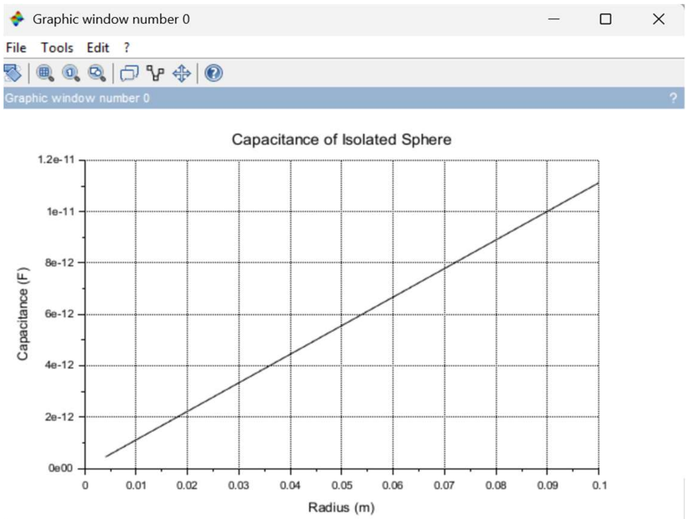
Scilab 2026.1.0 Console

```
Enter the inner radius r1 (m): 3*10^-3

Enter the outer radius r2 (m): 5*10^-3

  "Capacitance of isolated sphere (F):"
   5.563D-13
  "Capacitance of concentric spheres (F):"
   8.345D-13
exec: Wrong number of output argument(s): 0 expected.

-->
```



(7) Given

$r_1 = 3mm$

$r_2 = 5mm$

Formula

$$C_i = 4\pi\varepsilon_0 r_2$$

$$C_i = 4\times3.14\times8.854\times10^{-12}\times0.005$$

$$C_i = 12.56\times8.854\times10^{-12}\times0.005$$

$$C_i = 111.20824\times10^{-12}\times0.005$$

$$C_i = 0.5560412\times10^{-12}$$

$$\boxed{C_i = 5.56\times10^{-13}F}$$

O/P

Formula

$$C_o = \frac{4\pi\varepsilon_0 r_1 r_2}{r_2 - r_1}$$

$$C_o = \frac{4\times3.14\times8.854\times10^{-12}\times(0.003)(0.00}{0.005 - 0.003}$$

$$= \frac{12.56\times8.854\times10^{-12}\times15\times10^{-5}}{0.002}$$

$$= \frac{1.668\times10^{-15}}{0.002}$$

$$\boxed{C_o = 8.34\times10^{-13}F}$$

CASE 1:-CONCENTRIC SPHERICAL CAPACITOR WITH DIFFERENT DIELECTRIC MEDIUMS

7.1.sci (C:\Users\abhis\Documents\7.1.sci) - SciNotes

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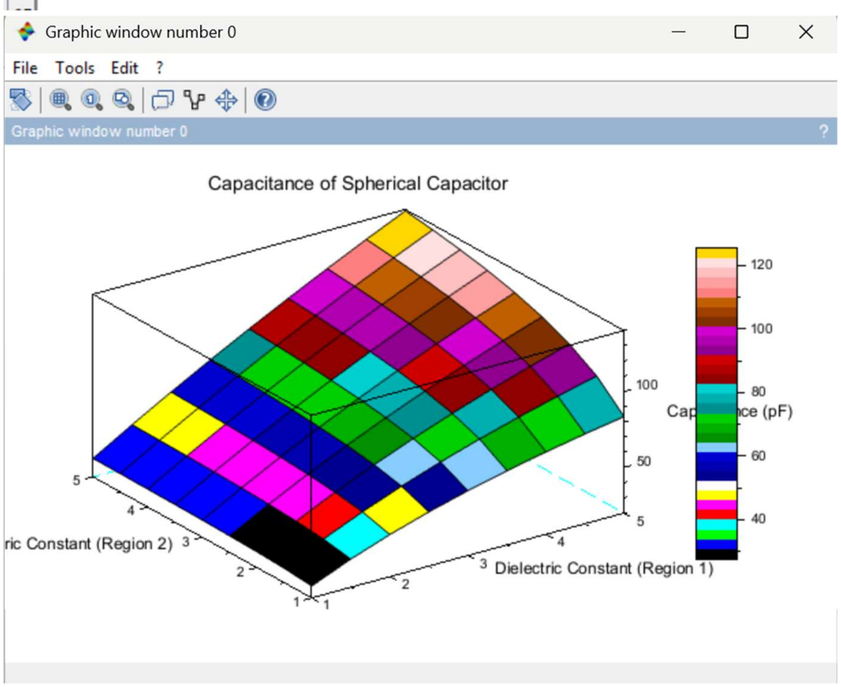
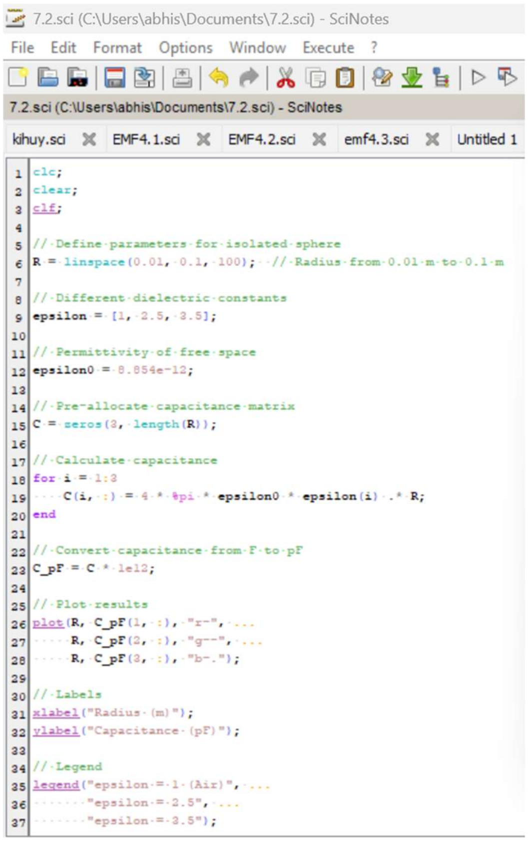
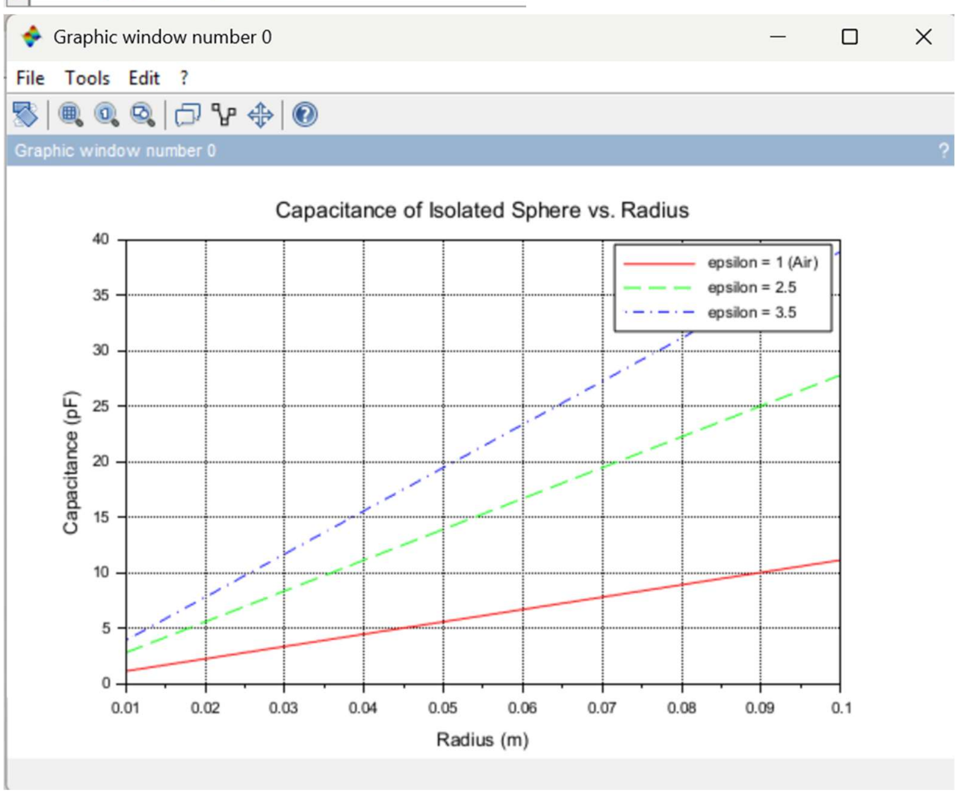
```
clc;
clear;
clf;

// Constants
epsilon_0 = 8.854e-12;

// Radii (m)
R1 = 0.2;
Rm = 0.6;
R2 = 1.0;

// Dielectric constants
epsilon_r1 = 1:0.5:5;
epsilon_r2 = 1:0.5:5;

// Initialize capacitance matrix
C = zeros(length(epsilon_r1), length(epsilon_r2));

// Capacitance calculation
for i = 1:length(epsilon_r1)
    for j = 1:length(epsilon_r2)

        // Capacitance of region 1
        C1 = 4 * %pi * epsilon_0 * epsilon_r1(i) ...
             * (R1 * Rm) / (Rm - R1);

        // Capacitance of region 2
        C2 = 4 * %pi * epsilon_0 * epsilon_r2(j) ...
             * (Rm * R2) / (R2 - Rm);

        // Equivalent capacitance
        C(i, j) = 1 / (1/C1 + 1/C2);

    end
end
```



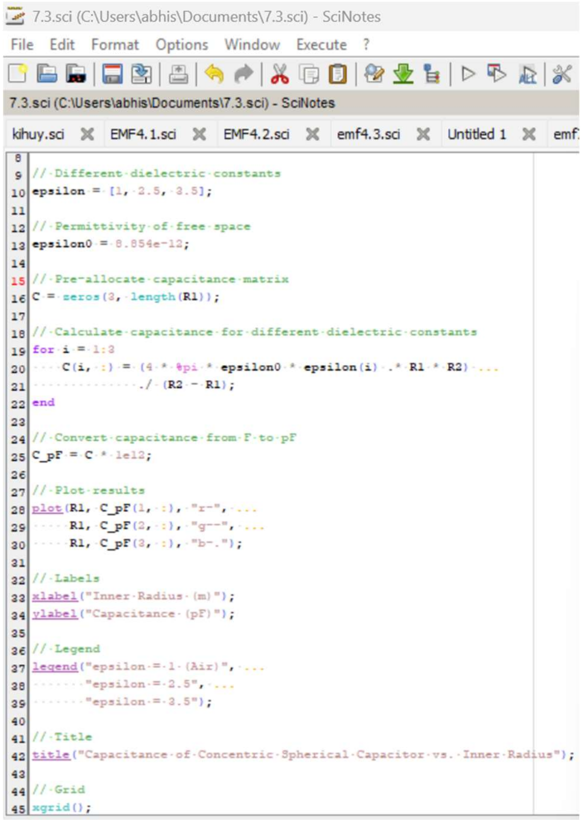
CASE 2: ISOLATED SPHERE




```
clc;
clear;
clf;

// Define parameters for isolated sphere
R = linspace(0.01, 0.1, 100);  // Radius from 0.01 m to 0.1 m

// Different dielectric constants
epsilon = [1, 2.5, 3.5];

// Permittivity of free space
epsilon0 = 8.854e-12;

// Pre-allocate capacitance matrix
C = zeros(3, length(R));

// Calculate capacitance
for i = 1:3
    C(i, :) = 4 * %pi * epsilon0 * epsilon(i) .* R;
end

// Convert capacitance from F to pF
C_pF = C * 1e12;

// Plot results
plot(R, C_pF(1, :), "r-", ...
     R, C_pF(2, :), "g--", ...
     R, C_pF(3, :), "b-.");

// Labels
xlabel("Radius (m)");
ylabel("Capacitance (pF)");

// Legend
legend("epsilon = 1 (Air)", ...
       "epsilon = 2.5", ...
       "epsilon = 3.5");
```



CASE 3: CONCENTRIC SPHERICAL CAPACITOR



7.3.sci (C:\Users\abhis\Documents\7.3.sci) - SciNotes

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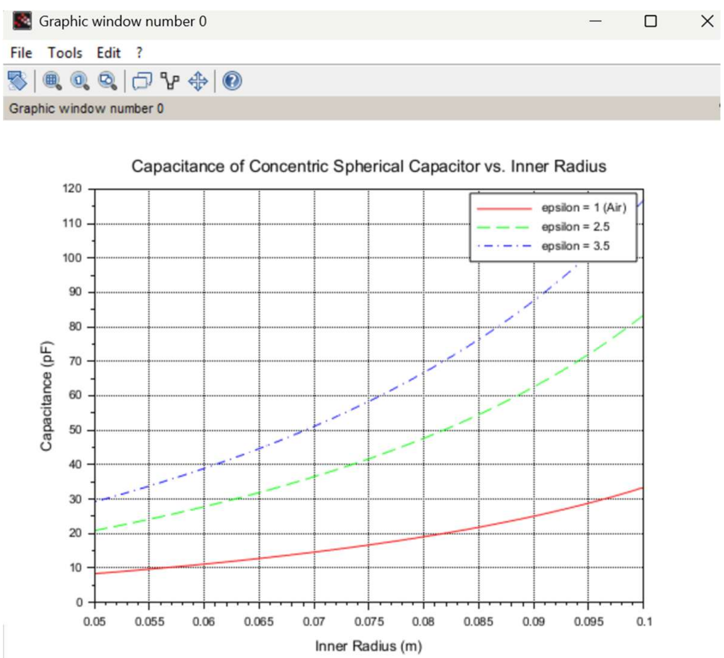
```
// Different dielectric constants
epsilon = [1, 2.5, 3.5];

// Permittivity of free space
epsilon0 = 8.854e-12;

// Pre-allocate capacitance matrix
C = zeros(3, length(R1));

// Calculate capacitance for different dielectric constants
for i = 1:3
    C(i, :) = (4 * %pi * epsilon0 * epsilon(i) .* R1 * R2) ...
              ./ (R2 - R1);
end

// Convert capacitance from F to pF
C_pF = C * 1e12;

// Plot results
plot(R1, C_pF(1, :), "r-", ...
     R1, C_pF(2, :), "g--", ...
     R1, C_pF(3, :), "b-.");

// Labels
xlabel("Inner Radius (m)");
ylabel("Capacitance (pF)");

// Legend
legend("epsilon = 1 (Air)", ...
       "epsilon = 2.5", ...
       "epsilon = 3.5");

// Title
title("Capacitance of Concentric Spherical Capacitor vs. Inner Radius");

// Grid
xgrid();
```



Result : Thus the program was verified for Capacitance of an isolated sphere and two concentric sphere using SCILAB was successfully.